Thermal physics

Igcse Physics

States of matter 物态

Matter exists in three states: **solid** 固体, **liquid** 液体 and **gas** 气体.

State	Shape	Volume	Particles
Solid	fixed	fixed	close together, in a regular pattern, vibrating
Liquid	takes the shape of the container	fixed	close together, no pattern, can move past each other
Gas	fills the container	changes	far apart, fast, random motion

The changes of state are: melting (solid → liquid), boiling/evaporating (liquid → gas), **condensation** 凝结 (gas → liquid) and **solidification** 凝固 (liquid → solid).

The kinetic particle model 分子动理论

All matter is made of tiny **particles** 粒子 that are always moving. This is the kinetic particle model.

- In a solid the particles only **vibrate** 振动 about fixed positions.
- In a liquid they are still close but can slide past each other.
- In a gas they are far apart and move quickly in random directions.

Temperature and particle energy

When you heat a substance, its particles move faster, so they have more **kinetic energy** 动能. **Temperature** 温度 is a measure of the average kinetic energy of the particles.

The lowest possible temperature is **absolute zero** 绝对零度, $-273\,°\mathrm{C}$. At this point the particles have the least possible energy.

Gas pressure

Gas particles hit the walls of their container. Each hit is a tiny push. The **pressure** 压强 of the gas is the total force of these hits per unit area.

- Heating a gas (at constant volume) makes particles move faster and hit harder and more often, so the pressure rises.
- Squeezing a gas into a smaller volume (at constant temperature) means more hits per second on each unit of area, so the pressure rises.

For a fixed mass of gas at constant temperature:

$$pV = \text{constant}$$

So if the volume halves, the pressure doubles.

Brownian motion

If you look at smoke in air under a microscope, you see tiny specks moving in a jerky, random way. This is **Brownian motion** 布朗运动. The specks are pushed by fast, invisible air particles hitting them. It is strong evidence for the kinetic particle model.

The kelvin scale

Scientists often use the **kelvin** 开尔文 (K) temperature scale, which starts at absolute zero. To convert:

$$T \text{ (in K)} = \theta \text{ (in °C)} + 273$$

So $0\,°\text{C} = 273$ K.

Thermal expansion 热膨胀

When matter is heated, its particles move more and take up more space, so the material expands. Gases expand the most, then liquids, then solids.

Everyday examples: gaps are left between railway lines; bridges sit on rollers; a tight metal lid loosens when heated.

Internal energy and specific heat capacity

The **internal energy** 内能 of an object is the total energy of all its particles. Heating an object raises its internal energy and usually its temperature.

The **specific heat capacity** 比热容 is the energy needed to raise the temperature of 1 kg of a material by 1 °C.

$$c = \frac{\Delta E}{m\,\Delta\theta}$$

A material with a high specific heat capacity (like water) needs a lot of energy to warm up and cools down slowly.

Melting, boiling and evaporation

Melting 熔化 and **boiling** 沸腾 need energy, but the temperature stays the same while the state changes. This energy breaks the forces between particles. For water at normal air pressure, melting is at $0\,°\text{C}$ and boiling at $100\,°\text{C}$.

Evaporation 蒸发 is when a liquid changes to a gas at its surface, below the boiling point. The fastest particles escape from the surface. Because the fastest (most energetic) particles leave, the average energy of those left behind falls, so the liquid **cools down** 冷却.

Evaporation is faster when the temperature is higher, the surface area is larger, and there is more air movement over the surface.

Transfer of thermal energy 热能传递

Thermal energy moves from hotter places to colder places in three ways.

Conduction

Conduction 热传导 is the transfer of thermal energy through a material without the material moving. Heated particles vibrate more and pass the energy to their neighbours. In metals, free (**delocalised**) **electrons** 自由电子 carry energy quickly, so metals are good **thermal conductors** 热导体.

Materials that conduct badly (like air, wood and plastic) are **thermal insulators** 热绝缘体.

Convection

Convection 对流 happens in liquids and gases. When a fluid is heated it expands, becomes less dense, and rises. Cooler, denser fluid sinks to take its place. This circle of moving fluid is a **convection current**.

Convection cannot happen in a solid because the particles cannot move from place to place.

Radiation

Thermal radiation 热辐射 is energy carried by **infrared** 红外线 waves. All objects emit it, and it needs no material to travel through —it can cross empty space (this is how energy reaches us from the Sun).

A surface that is **dull** 暗淡 and black is a good **emitter** 发射体 and a good **absorber** 吸收体 of infrared. A surface that is shiny and white is a poor emitter and a good **reflector** 反射体.

An object stays at a constant temperature when it emits energy at the same rate as it absorbs energy. The rate of emission is greater when the surface is hotter and larger.